# It's Hard: Why the Physics Community Will Never Close Its Open Problems

## The Structural Architecture of Institutional Non-Closure

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## 1. The Phrase

Physics has a phrase it reaches for when its open problems remain open. The phrase is "it's hard."

Unifying gravity with quantum mechanics is hard. Explaining why the cosmological constant is small is hard. Measuring the gravitational constant precisely is hard. Understanding why certain parameters are zero is hard. The phrase appears in grant applications, in public lectures, in responses to outside questions, and in the introductions of papers that propose new approaches to old problems. It is delivered with the authority of people who work with the fundamental constituents of reality. The implication is clear: the problem is hard, the people working on it are the best qualified to assess its difficulty, and if it hasn't been solved yet, that tells you about the depth of the problem, not about the competence of the community.

In any engineering discipline, the phrase would not survive first contact with a project review. If an engineer said "the sensors don't converge, and it's hard," the project manager would ask: what have you done to identify the source of the non-convergence? What variables have you correlated against? What's your instrumentation plan? "It's hard" is not a status report. It's the absence of one. It tells you nothing about what has been tried, what has been ruled out, what the next step is, or when the problem will be closed. It functions as a request to stop asking.

This paper makes one claim: the major open problems in physics are not hard. They are kept open by identifiable structural features of the institution that practices physics. Each feature is mechanical. Each can be described without invoking difficulty. Each prevents closure in a specific, traceable way. The paper will identify each feature, demonstrate its effect, and commit to falsification conditions that would prove the analysis wrong.

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## 2. The Open Problems

Physics maintains a list of open problems. The list has been broadly stable for decades. Each problem has generated thousands of papers, hundreds of PhD theses, and billions of dollars in research funding. Each remains open.

**The strong CP problem.** In the theory that describes the strong nuclear force — quantum chromodynamics — there is a parameter called theta. Theta controls the degree to which certain symmetries are violated. Theta could take any value from zero to $2\pi$. Every measurement of theta returns zero. The problem, as the community defines it, is: why is theta zero? The problem has been open since the 1970s. The leading proposed solution is a hypothetical particle called the axion, which has not been detected.

**The gravitational constant non-convergence.** The gravitational constant G describes how strongly masses attract each other. G has been measured by independent experiments around the world for over two centuries. The independent measurements do not agree. The disagreement is larger than the individual experiments' stated measurement uncertainties. This means either the uncertainty estimates are wrong, the instruments are sensitive to something unidentified, or G is not what the community thinks it is. The problem has been open since precision measurements began revealing the disagreement in the late 20th century. The institutional response has been to compute a weighted average of the non-convergent measurements and publish it as the recommended value.

**The cosmological constant problem.** Quantum field theory predicts a vacuum energy density. The prediction is larger than the observed value by a factor of roughly $10^{54}$ to $10^{120}$, depending on the calculation method. This is sometimes called the worst prediction in physics. The problem has been open since the 1980s. No resolution is in sight.

**Quantum gravity.** General relativity describes gravity as the geometry of spacetime. Quantum mechanics describes particles and forces as quantized fields. The two frameworks are mathematically incompatible — general relativity is a smooth, continuous theory, and quantum mechanics is a discrete theory. Combining them has been the central goal of theoretical physics for most of a century. String theory, loop quantum gravity, and other approaches have been proposed. None has produced a testable prediction confirmed by experiment.

**The hierarchy problem.** The Higgs boson's measured mass is roughly 125 GeV. Quantum corrections to the Higgs mass should push it up to the Planck scale — roughly $10^{19}$ GeV — unless the corrections cancel with extraordinary precision. Why the Higgs mass is so much smaller than the Planck scale is the hierarchy problem. Proposed solutions include supersymmetry, which predicts partner particles for every known particle. No partner particle has been detected. The problem has been open since the 1970s.

**Dark matter and dark energy.** Approximately 27% of the universe's energy content is attributed to dark matter — matter that interacts gravitationally but has not been directly detected. Approximately 68% is attributed to dark energy — a component that drives the accelerating expansion of the universe and whose nature is unknown. Together, 95% of the universe's content is named but unidentified. Multiple detection experiments for dark matter particles have run for decades. None has produced a confirmed detection.

Each of these problems has generated extensive research programs. Each has consumed substantial public funding over multiple decades. Each remains open. The community's explanation for why they remain open is that they are hard. This paper offers a different explanation.

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## 3. Two Kinds of Claim

Physics carries two categories of claim simultaneously and treats them as one enterprise. The distinction between them is structural, and the failure to maintain the distinction is the root cause of the institutional dysfunction this paper describes.

The first category is precision measurement — values confirmed by operational systems to extraordinary accuracy. The electron's anomalous magnetic moment has been measured and predicted to better than one part in a trillion. Fifteen significant digits of agreement between theory and experiment. The GPS satellite system applies relativistic corrections to its atomic clocks every second of every day, and the corrections work at nanosecond precision — if they were wrong, your phone would locate you in the wrong city. The primordial abundances of light elements produced in the first minutes after the Big Bang have been predicted from nuclear physics and confirmed by satellite observations of the cosmic microwave background and spectroscopic measurements of distant gas clouds. These are not hypotheses. They are operational facts that systems depend on.

The second category is interpretive framework — conceptual commitments that have never been subjected to a mechanical test. The block universe, which asserts that past, present, and future all exist simultaneously as a four-dimensional structure. The treatment of spacetime as a single unified substance. The treatment of mass as a form of stuff rather than a measure of resistance. The treatment of continuous mathematics as the mechanism of physical reality rather than an approximation of it. These are not measurements. They are interpretations. They are taught alongside the measurements in the same courses, published in the same journals, and funded by the same grants. A student learning physics absorbs both and has no structural way to distinguish which claims are backed by 15-digit measurement precision and which are conceptual traditions that have never faced a falsification test.

Consider a workbench. A precision engineer assembling a delicate mechanism needs a clean surface. Every tool that isn't being used, every part that doesn't belong to the current assembly, every piece of scrap from a previous project — each is an obstacle.

The engineer clears the bench before beginning. Only the parts and tools needed for the current work remain.

The physics workbench has never been cleared. Precision measurements confirmed at parts per trillion sit next to unfalsified interpretive commitments from the early 20th century. Every time a new measurement needs to be reconciled with the existing framework, it must be made compatible with everything on the bench — both the confirmed measurements and the untested interpretations. When reconciliation fails, the failure is called a problem, a mystery, a tension. The problems accumulate because the bench accumulates. New measurements add items. Nothing is ever removed. The "hard problems" are, in significant part, the difficulty of reconciling real measurements with framework commitments that should have been cleared away decades ago.

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## 4. The Concrete Won't Dry

The gravitational constant G is the clearest example of a measurement problem being held open by institutional structure.

G describes the strength of gravitational attraction between masses. It appears in Newton's law of gravitation — the force between two masses is proportional to G times the product of the masses divided by the square of the distance between them. G has been measured since Henry Cavendish's 1798 experiment using a torsion balance. In the two centuries since, multiple independent laboratories have measured G using different apparatus — torsion balances, beam balances, atom interferometers, and other techniques.

The measurements do not converge. The values produced by different laboratories, using different instruments, disagree with each other by amounts larger than the laboratories' own stated uncertainties. When a laboratory says "we measured G to be this value, and we are confident our measurement is accurate to within this uncertainty range," the values from other laboratories fall outside that range. This is not a matter of one or two outliers. The non-convergence is systematic and persistent across decades of independent measurement.

In any engineering context, non-convergent sensor readings are the most important signal in the system. If three temperature sensors in the same room read 70°F, 73°F, and 68°F, and each claims accuracy to $\pm$ 0.5°F, the readings are incompatible. The first step is not to average them. The first step is to investigate why they disagree. Is one sensor miscalibrated? Are the sensors in different locations with different airflow? Is the room temperature actually non-uniform? Each explanation is testable. Each test narrows the search space. The engineer investigates until the non-convergence is explained.

The Committee on Data for Science and Technology — CODATA — takes the opposite approach with G. Every several years, the committee collects the non-convergent measurements, assigns weights based on its assessment of each measurement's reliability, computes a weighted average, and publishes the average as the recommended value of G. The recommended value shifts from cycle to cycle. The uncertainty band is adjusted —

typically widened — to accommodate the spread. The non-convergence is noted in the committee's reports. It is not resolved.

The recommended value is the number that propagates. Every gravitational calculation performed by every physicist and engineer worldwide uses the recommended value. Every orbital prediction, satellite trajectory, cosmological model, and mass determination uses a number that was produced by averaging measurements that don't agree with each other. If G depends on something that differs between the measurement contexts, every downstream calculation inherits an error that cannot be traced because the structure of the non-convergence was destroyed by the averaging.

Now consider the scope of the measurements. Every measurement of G in the history of physics has been taken on the surface of the Earth. The altitude varies by a few kilometers at most. The gravitational potential is essentially the same at every measurement site. The geological environment varies, but within the narrow range of Earth's crustal composition. Nobody has measured G at the Earth-Sun Lagrange points. Nobody has measured G in deep space. Nobody has measured G on the Moon, on Mars, in orbit, or at any gravitational potential substantially different from Earth's surface.

The field calls G a universal constant. Universal means it applies everywhere — at every location, at every scale, in every gravitational environment, at all times throughout the history of the universe. This claim is based on measurements taken within a thin shell on the surface of one planet, and those measurements don't even agree with each other within that shell.

Compare with the electron's anomalous magnetic moment. Measured to 15 significant digits. Theory and experiment agree. The measurement is of the electron itself — its intrinsic properties — and electrons are identical everywhere. The measurement can legitimately claim universal validity because it measures a property of a fundamental particle, not a coupling that might depend on environment.

G measures a coupling between masses. Every measurement of that coupling is embedded in a massive environment — the Earth — that cannot be removed from the experiment. If the coupling depends on the surrounding gravitational environment, the dependence is invisible because every measurement is taken in essentially the same environment. The non-convergence might be the first sign that the coupling isn't constant — and averaging the non-convergence away ensures the sign is never read.

A bridge engineer whose concrete doesn't cure to specification doesn't average the test results and proceed with construction. The engineer stops building and investigates the concrete. The engineer has professional accountability for what happens when traffic crosses the bridge. Physics has no equivalent accountability for what happens when a non-convergent constant propagates into every gravitational calculation on the planet.

The non-convergence of G is not hard. It is uninvestigated. The data that would enable investigation — time-stamped readings from every instrument with full metadata about device type, location, altitude, geological environment, and operating conditions — exists in laboratory archives. It is not published. It is not in a standardized queryable format. It is not available for cross-laboratory comparison. The infrastructure to publish and query it is commodity technology used daily by industries that monitor thousands of sensors simultaneously. The institutional decision to publish an average instead of the raw data is a choice. The choice prevents investigation. The investigation would resolve the non-convergence. The resolution might require updating the model. The model has not been updated. “It’s hard.”

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## 5. Accepting Everything Except Zero

The strong nuclear force is described by quantum chromodynamics — QCD. The theory contains a parameter called theta that governs the degree to which certain symmetries are violated in strong-force interactions. The parameter is continuous — it can take any value from zero to $2\pi$. Every measurement of theta, through experiments sensitive to the symmetry violations it would produce, returns zero or a value consistent with zero within experimental uncertainty.

Zero is the measured value. The field does not accept it.

The field says: theta could be any value from zero to $2\pi$. The fact that it is zero requires an explanation. Why zero and not some other value? This question — why is a measured value what it is — has generated an entire subfield. The leading proposed explanation is the Peccei-Quinn mechanism, which introduces a new symmetry and a new particle, the axion, that dynamically drives theta to zero. The axion has been searched for experimentally for decades. It has not been found. The search continues because the community considers it more plausible that an undiscovered particle drives theta to zero than that theta is simply zero.

Now consider the fine structure constant, alpha. Alpha describes the strength of the electromagnetic force. At low energies, alpha is approximately 1/137. This is a measured value. Nobody says: alpha could be any value, so why is it 1/137? Nobody has invented a particle to explain why alpha has its particular value. It is measured, recorded, and used.

But alpha does something more dramatic than theta. Alpha runs — it changes value depending on the energy scale at which it is measured. At the scale of the Z boson, approximately 91 GeV, alpha is approximately 1/128. At higher energies approaching the grand unification scale, alpha is projected to be roughly 1/42. A quantity called a “constant” varies by a factor of more than three across accessible energy scales. This is accepted without existential crisis. No new particle is needed to explain why alpha runs. No subfield has been created around the alpha-running problem.

The asymmetry is stark. A parameter measured at zero, consistently, across every experiment, is treated as a mystery requiring new physics. A parameter that changes by 300% across energy scales is called a constant and accepted as given. The selection criterion is not scientific. A scientific criterion would treat both measured values with equal seriousness. The actual criterion is institutional: zero threatens the framework — it implies something structural about the strong force that the current formalism does not explain — while a running coupling constant is accommodated by the formalism through well-understood renormalization group equations.

The community's posture toward theta = 0 is the inverse of the scientific method. The scientific method says: make a measurement, accept the result, update the model. The community's posture says: make a measurement, reject the result as coincidental, invent a mechanism to explain why the measurement accidentally gives this particular value, and search for the mechanism experimentally. This is not a caricature. This is the actual structure of the strong CP research program. The measurement says zero. The program says: zero is suspicious, here's why it might secretly be driven to zero by an undiscovered particle.

If the field applied the same standard to theta that it applies to the electron's magnetic moment — measure it, accept the value, compute with it — the strong CP problem would not exist. It exists because the community treats zero as more suspicious than any other value, which is a philosophical position, not a scientific one. Zero is a measurement result. It is the most precisely constrained measurement result in the strong-force sector. The field should use it.

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## 6. Teaching Without Mechanism

The block universe is the assertion that past, present, and future all exist simultaneously. All of spacetime — every event that has ever occurred and every event that ever will occur — exists as a four-dimensional structure. Time does not flow. Nothing changes. The appearance of change is an artifact of consciousness scanning through a pre-existing block. The present moment has no special status. Your birth, your death, and every moment between them coexist in the block right now, along with the formation of the Earth and the heat death of the universe.

This is taught in universities. It is discussed in general relativity textbooks. It is presented at academic conferences. It is treated as a legitimate interpretation of the mathematical formalism of Minkowski spacetime, which represents events as points in a four-dimensional space with a specific geometric structure.

An engineer hearing this description asks one question: where is it?

The question is not philosophical. It is physical. The block universe makes a physical claim — that states exist. Existence requires physical instantiation. Physical instantiation requires a substrate, energy, and a mechanism.

Consider the storage requirement. The observable universe contains approximately $10^{80}$ particles. The universe is approximately $10^{61}$ Planck times old, where the Planck time — roughly $5.4 \times 10^{-44}$ seconds — is the smallest physically meaningful time interval. If every state of every particle at every Planck time exists simultaneously in the block, the block contains on the order of $10^{141}$ state entries. This number does not include the spatial resolution needed at each time step, which would multiply it enormously. Where does this information sit? What physical substrate stores it? What energy source maintains it?

Consider the thermodynamic problem. If all states coexist, the entropy of the block is fixed. Entropy does not increase because nothing changes — all states are already present.

But entropy increase is one of the most robustly confirmed facts in physics. Heat flows from hot to cold. Stars burn fuel irreversibly. Chemical reactions proceed in preferred directions. Eggs break and do not spontaneously reassemble. These are not perceptual artifacts. They are energy transfers with measurable physical consequences. The block universe requires the second law of thermodynamics to be an illusion — your consciousness merely perceives entropy increasing as it scans through the block. Decades of experimental thermodynamics say otherwise.

Consider the heat signature. Storing information requires energy. This is not a theoretical claim — it is a consequence of Landauer's principle, which is experimentally confirmed. Maintaining $10^{141}$ state entries requires energy proportional to the storage. That energy would have thermodynamic signatures. It would radiate. It would be detectable. It is not detected. No observation in the history of physics is consistent with a storage substrate maintaining the block universe.

The block universe makes physical claims. The physical claims have testable consequences. The consequences are not observed. By any standard of falsification, the block universe is dead. It persists because the physics community treats "interpretations" as a special category exempt from falsification. The exemption has no scientific basis. An interpretation that asserts physical existence is a physical claim. Physical claims must be tested. Untested physical claims must be removed from the workbench, or they clutter every subsequent attempt at assembly.

A computational model does not have this problem. A processor maintains current state — the values in its registers at this moment. The processor applies an update rule. The registers change. The previous state is overwritten. The next state does not exist until the update executes. Storage requirement: one state, not $10^{141}$. Energy requirement: the energy to perform one update, not the energy to maintain all updates simultaneously. No block. No substrate storing the past. No pre-existing future. Current state, update rule, next state. This is how every real computational system works. Nobody building a computer proposes that all past and future register states exist simultaneously. The proposal would be absurd. The storage and energy requirements would be physically impossible. The universe is a physical system. The same constraints apply.

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## 7. Spacetime Decomposes — GPS Proves It

The Global Positioning System is a constellation of satellites carrying atomic clocks. Your position is determined by comparing the time signals from multiple satellites. For this to work, the clocks must be synchronized to nanosecond precision. They cannot be simply set to the same time and left alone, because two relativistic effects cause them to drift.

The first effect is gravitational time dilation. A clock at higher altitude — farther from the Earth's center, in a weaker gravitational field — runs faster than a clock at the surface. GPS satellites orbit at approximately 20,200 kilometers altitude. At this altitude, the gravitational time dilation causes the satellite clocks to gain approximately 45 microseconds per day relative to ground clocks.

The second effect is velocity time dilation. A clock in motion runs slower than a stationary clock. GPS satellites travel at approximately 3,870 meters per second. At this velocity, the velocity time dilation causes the satellite clocks to lose approximately 7 microseconds per day relative to ground clocks.

The net effect is a gain of approximately 38 microseconds per day. The gravitational contribution is approximately 86% of the total correction magnitude. The velocity contribution is approximately 14%. These two contributions are computed separately, using different formulas drawn from different physical mechanisms, and applied as separate corrections. Every GPS receiver on the planet performs this decomposition. The system has operated continuously since the 1990s. The corrections work.

The formalism of general relativity describes spacetime as a single four-dimensional manifold with a metric that encodes both spatial and temporal relationships. The formalism says: spacetime is one thing. Space and time are not separate — they are aspects of a single geometric structure.

The engineering says: compute two separate corrections or the system doesn't work. The gravitational correction comes from general relativity's treatment of clocks in gravitational potentials. The velocity correction comes from special relativity's treatment of moving clocks. They involve different physics, different magnitudes, and different mathematical operations. They are not interchangeable aspects of one calculation. They are two calculations that produce two numbers that are added together.

If spacetime were operationally one thing, one unified calculation would produce the total correction. The fact that the correction must be decomposed into a spatial-gravitational component and a temporal-velocity component, with a measured ratio of roughly 86:14, demonstrates that space and time contribute differently and separably. The unification is a property of the formalism, not of the physics. The formalism is mathematically convenient. The engineering is operationally necessary.

This does not mean general relativity is wrong. General relativity's predictions are confirmed by GPS to extraordinary precision — that is the point. The predictions work. But they work by decomposition, not by unification. The theory predicts correctly. The interpretation that spacetime is a single substance is an additional claim beyond the predictions. The additional claim is not required by the mathematics, is not required by the engineering, and is contradicted by the operational decomposition that every GPS receiver performs every second.

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## 8. Mass Is Resistance

Newton's second law: F = ma. Force equals mass times acceleration. Rearrange it: m = F/a. Mass equals force divided by acceleration. Mass is the ratio of applied force to resulting acceleration. In operational terms, mass is how much something resists being pushed. Push something and it barely moves — high mass. Push something and it flies — low mass. Mass is resistance to acceleration. That is its operational definition. That is what every measurement of mass actually measures.

The Higgs mechanism, confirmed by the discovery of the Higgs boson in 2012, explains how fundamental particles acquire their mass. The explanation is precise: particles interact with the Higgs field. The strength of the interaction — the coupling constant — determines how much the particle resists acceleration. A particle with strong Higgs coupling resists acceleration more and is measured as having more mass. A particle with zero Higgs coupling — the photon — has zero resistance to acceleration and travels at the speed of light.

The Higgs field does not fill particles with stuff. It does not "grant substance." It determines a coupling strength that produces resistance to acceleration. That is inertia. Mass is inertia. The Higgs mechanism is an inertia mechanism. The physics is clear on this.

The language is not. Popular descriptions say the Higgs field "gives particles their mass," as though mass is a substance being distributed. Textbooks speak of mass as a property that particles "have," like a possession. The metaphor of the Higgs field as molasses that particles move through reinforces the intuition of mass as stuff — some medium that creates drag, filling or coating particles with heaviness.

The difference between substance and resistance is not semantic. It determines which questions you ask and which research programs you pursue. If mass is a substance, you ask: what is it made of? How much of it is there? Where is it located? If mass is resistance, you ask: what determines the coupling? How does the coupling vary? What is the coupling sensitive to? These are different questions leading to different experiments.

The equivalence principle — one of the deepest facts in physics — states that inertial mass and gravitational mass are equal. Inertial mass is how much something resists acceleration. Gravitational mass is how strongly something couples to the gravitational field. Every experiment ever conducted confirms they are identical.

If mass is substance, the equivalence is a coincidence that demands explanation. Why should the amount of stuff in an object be exactly equal to how strongly the object interacts with gravity? This coincidence has generated decades of theoretical work.

If mass is resistance, the equivalence is a tautology. Resistance to acceleration and coupling to the gravitational field are the same phenomenon measured two different ways. There is nothing to explain. The "coincidence" dissolves because there was never a coincidence — there was a single phenomenon with two names.

The two-name problem runs deeper than terminology. The Newtonian formalism has two separate slots where mass appears. Gravitational mass in $F = GMm/r^2$. Inertial mass in $F = ma$. Einstein elevated their equality to an axiom — the equivalence principle — and built general relativity on it. But the reason it needs to be an axiom is that the formalism has two slots. If the formalism used one concept — resistance — there would be one slot, the equivalence would be built into the structure, and the axiom would be unnecessary.

Every sensory experience that creates the perception of substance is actually a field interaction. Vision is photon exchange — photons from a light source interact with electron energy states in an object's atoms, and the photons that reach your retina trigger chemical responses in your photoreceptor cells. Color is a frequency. Shape is a boundary pattern.

Touch is electromagnetic repulsion — your hand's electron shells repel the object's electron shells. You never make contact with anything. Sound is pressure waves in molecules held apart by electromagnetic forces. Every sensory channel detects field interactions. No sensory channel detects substance. Substance is a perceptual compression that human nervous systems generate for survival efficiency. It is not physics.

Physics knows this. Quantum field theory describes every particle as an excitation of a field, not as a piece of stuff. The electron is an excitation of the electron field. The photon is an excitation of the electromagnetic field. None of them are substance. The knowledge is explicit in the formalism and ignored in the language. The language shapes intuition. Intuition shapes the research program. The research program spends decades trying to unify quantities that were never separate — they were named separately by people in different centuries solving different problems.

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## 9. The Universe Is Discrete and Finite

The Planck time is approximately $5.4 \times 10^{-44}$ seconds. It is the smallest time interval at which the concept of sequential time retains physical meaning. Below this interval, the known laws of physics do not produce meaningful predictions. The Planck length is approximately $1.6 \times 10^{-35}$ meters — the corresponding smallest meaningful spatial interval.

These are not approximate values of quantities that might be smaller. They are structural limits. They emerge from the three fundamental constants that define the boundary between quantum mechanics and gravity: the gravitational constant G, the reduced Planck constant $\hbar$, and the speed of light c. The Planck scale is where quantum mechanics and gravity meet. Below it, the distinction between spacetime and quantum fluctuations dissolves.

If the universe has a minimum time interval, the universe operates on a clock. Each Planck time interval is a tick. At each tick, the state updates. The update is finite — a finite number of operations on a finite state producing a finite output. The universe advances one tick. The state updates. The universe advances another tick. This is what a discrete computational system does. It is how every processor operates, how every digital simulation operates, how every finite state machine operates.

A finite operation cannot compute an infinite decimal. The number $\pi$ has an infinite decimal expansion — 3.14159… continuing without end. No finite computation can produce all of those digits. A finite computation can produce any specific number of digits, but not infinitely many. If the universe updates state using finite operations on each Planck tick, the universe does not use real numbers. Real numbers — the mathematical objects with infinite decimal expansions that form the continuous number line — are a mathematical abstraction. They are not computed by the physical universe because the physical universe does not have infinite time within a single tick to compute them.

Every equation in physics that uses real-number arithmetic — which is essentially every equation in physics — is therefore an approximation. The approximation works extraordinarily well at scales much larger than the Planck scale, in the same way that smooth curves drawn through discrete data points work extraordinarily well when the points are close together. But the smooth curve is not the data. The data is the discrete points. The continuous mathematics is not the mechanism. The mechanism is the discrete update. The continuous equations are effective descriptions that emerge from the discrete process in the limit of many ticks and many Planck cells. They are not the process itself.

This has a direct consequence for the infinities that appear in physics. Quantum field theory, when calculating the interactions between particles, routinely produces infinite answers. These infinities are removed through a procedure called renormalization — essentially, subtracting one infinity from another to produce a finite result. The procedure works. The finite results match experiment to extraordinary precision. But the infinities were never physical. They are artifacts of applying continuous mathematics to a system that operates discretely. The universe never produces an infinity. It never computes an infinite sum. It never evaluates a continuous integral. It performs finite operations per tick and moves on. The infinities exist in the formalism, not in the physics.

The observable universe contains a finite number of Planck volumes. Current estimates put the number at approximately $10^{184}$. That is a large number. It is not infinity. The state space of the universe is finite. The number of distinct configurations the universe can take is finite. The update rule operates on finite state. The output at each tick is finite. Nothing is infinite anywhere in the physical universe.

Infinity is a mathematical tool. It is a useful tool — it enables calculus, continuous analysis, and the powerful machinery of differential equations that has been the language of physics for centuries. But useful tools are not necessarily accurate descriptions of the system they are applied to. A smooth curve fitted to discrete data is a useful tool. It is not the data. Continuous mathematics applied to a discrete universe is a useful tool. It is not the universe.

The mismatch between continuous formalism and discrete reality is not an abstraction. It has concrete consequences. The quantum gravity problem — combining general relativity with quantum mechanics — is a problem of combining a continuous theory with a discrete theory. If the universe is discrete at the Planck scale, the continuous theory is the approximation and the discrete theory is the reality. The "problem" of quantum gravity is, in part, the problem of taking a continuous approximation too seriously and then struggling to reconcile it with the discrete structure it approximates. An engineer approaching this problem would start from the discrete system and derive the continuous behavior as an emergent property. Physics has historically done the reverse — starting from the continuous formalism and trying to discretize it. The direction matters. Discretizing a continuum introduces ambiguities. Deriving continuum behavior from a discrete system does not.

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## 10. The Data Destruction Pipeline

The Committee on Data for Science and Technology — CODATA — maintains the internationally accepted values of the fundamental physical constants. The values are updated on a cycle of several years. Each cycle, the committee collects measurements from laboratories worldwide, evaluates them, and publishes a set of recommended values.

The process begins with instruments. Each instrument measures a physical constant. Each instrument produces readings over time — a time series of measured values. Each instrument operates in a specific physical context — a specific location on the Earth's surface, at a specific altitude, in a specific geological environment, in a specific electromagnetic background, at specific times. The time series and the context together contain the complete record of what the instrument observed, when, and under what conditions.

This complete record is not what CODATA publishes.

CODATA reduces each instrument's time series to a summary statistic — an average with an uncertainty. The summaries from multiple instruments are collected. The committee assigns weights based on its assessment of each measurement's quality. A weighted average is computed across all instruments. The weighted average is published as the recommended value, accompanied by an uncertainty that reflects the committee's assessment of the combined evidence.

The recommended value is one number. The time series that went into it contained temporal structure — variations over hours, days, seasons, years. The temporal structure is gone. The individual instruments produced different readings — disagreement structure that could correlate with device type, location, altitude, or operating conditions. The disagreement structure is gone. Each measurement occurred in a physical context — environmental conditions that might affect the reading. The contextual information is gone.

Three categories of information that existed in the raw data have been destroyed: temporal structure, disagreement structure, and contextual correlation. The destruction is not accidental. It is the methodology. The methodology was designed in an era when collecting, storing, and distributing large datasets was expensive and technically impractical. Averaging was the most useful thing the available infrastructure could deliver. The infrastructure constraints no longer exist. Global time synchronization at nanosecond precision is available via GPS. Time series databases handling billions of data points are commodity technology. Standardized data formats for timestamped measurements are mature and widely deployed. Every component needed to publish the raw time series instead of the average exists, is tested, and is affordable. A single precision measurement experiment costs more than the entire data infrastructure that would preserve the raw structure.

The institutional decision to continue publishing averages instead of raw time series is a choice to continue destroying information that the public paid to generate. The instruments are publicly funded. The laboratories are publicly funded. The committee is publicly funded. The data belongs to the public. The public receives one number per constant per cycle. The time series that would enable independent analysis of temporal variation, instrument disagreement, and contextual effects sits in laboratory archives, unpublished, in non-standard formats, inaccessible.

If any fundamental constant depends on time, location, or context — even weakly — the averaging methodology ensures the dependence is never detected. The methodology was designed to assume constants are constant and to produce the best estimate of that constant value. The assumption is built into the processing. The data that would test the assumption is destroyed by the processing. The assumption cannot be challenged because the evidence needed to challenge it does not survive the pipeline.

The proposal to fix this is straightforward and is described in detail in [HOWL-CULT-13-2026]. Time-synchronize all contributing instruments to a common reference clock. Publish each instrument's readings as a time series with full metadata. Store the unified dataset in a queryable standardized format. The committee's role changes from producing the answer to maintaining the infrastructure and certifying data quality. The average can still be computed by anyone who wants it. But the raw structure is preserved. Nothing is destroyed.

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## 11. The Credential Gate

When a person outside the institutional credentialing system — someone without the expected doctoral degree, institutional affiliation, and publication history — identifies one of the structural problems described in this paper and submits work addressing it, the work is not evaluated on its content.

The first evaluation is credential-based. Does the author hold a doctorate in the relevant field? Are they affiliated with a recognized institution? Have they published in peer-reviewed journals? These questions are about the author. They tell you about the author's institutional history. They tell you nothing about whether the work is logically consistent, empirically grounded, mathematically correct, or useful.

If the author does not pass the credential check, the work is classified as outside the scope of serious consideration and excluded. The classification happens before the content is examined. Whether the work contains valid reasoning, references real measurements, performs correct derivations, or proposes testable predictions — none of these questions are asked. The classification intercepted the work before the questions could be reached.

The submitter receives a rejection. The rejection does not explain where the work fails in terms of content. It does not identify logical errors. It does not point to disagreements between the work's claims and established measurements. It does not locate mathematical mistakes. It provides no specific information the submitter can act on. The submitter cannot improve because they received no feedback about what specifically is wrong.

The perspectives most likely to see the structural problems described in this paper are precisely the perspectives that the credential gate excludes. A systems engineer who builds systems that update state on clock ticks looks at the Planck time and sees a clock rate for a finite state machine. A physicist trained in continuous field theory sees a minimum measurable interval and keeps writing integrals. A software architect who practices daily commitment to specific values with explicit failure conditions brings falsification as a lived engineering practice. A control theorist who models systems with feedback, stability, and convergence criteria brings frameworks that physics has never applied to its own measurement systems. These perspectives arise from training that physics departments do not provide and cannot generate internally.

The credential gate ensures that the field receives input only from people trained within its own pipeline. The pipeline teaches the existing frameworks — including the unfalsified interpretive commitments, the substance-based language, the continuous-mathematics-as-mechanism assumption, and the departmental boundaries that prevent cross-domain analysis. People who emerge from the pipeline share its conceptual commitments. They are unlikely to identify structural problems created by those commitments because the commitments are the foundation of their training. The people most likely to see the problems are trained elsewhere. The gate excludes them.

A content-based evaluation framework — evaluating work on its logic, its empirical grounding, its mathematical correctness, and its utility, regardless of who wrote it — is described in [HOWL-CULT-14-2026]. The framework does not guarantee that outsider contributions will be valuable. Most will fail content evaluation, and the failure patterns themselves are informative. But the current system guarantees that no outsider contribution, regardless of its content, will be examined. The guarantee is structural. It is built into the evaluation pipeline. The first filter is credential-based. Content evaluation never occurs.

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## 12. The Architecture of Non-Closure

Each structural feature described in this paper individually impedes progress toward closing the open problems. Together, they form a self-reinforcing system that cannot close its own problems because closing them would require changing the architecture, and the architecture is maintained by the people who would have to change it.

The data destruction pipeline removes the information needed to investigate non-convergence. The G measurements disagree. The disagreement might correlate with instrument type, location, altitude, geology, time, or environmental conditions. The correlation cannot be detected because the data was averaged before the correlation could be tested. To investigate the non-convergence, the field would need the raw data. To get the raw data, the field would need to change the data pipeline. The pipeline is maintained by the committee. The committee's role is to produce the recommended value. Changing the pipeline changes the committee's role. The committee does not change the pipeline.

The credential gate excludes the perspectives needed to see structural blind spots. The problems that physics cannot solve internally — unifying quantities that were never separate, reconciling continuous formalism with discrete reality, applying systems thinking to measurement infrastructure — require perspectives that physics training does not produce. Those perspectives exist in engineering, computer science, information theory, and control theory. The credential gate prevents those perspectives from entering the evaluation process. The gate is maintained by the credentialed community. The community sees no value in outsider contributions because it has never examined them. It has never examined them because of the gate. The gate is not challenged because the people inside the gate believe it is working.

The unfalsified frameworks clutter the workbench. Every new measurement must be reconciled with every item on the bench — confirmed measurements and untested interpretations alike. When reconciliation fails, the failure is called a new problem. The new problem generates research funding. Research funding supports careers. Careers depend on the continued existence of open problems. Clearing the bench — removing the unfalsified frameworks — would simplify the reconciliation landscape, potentially closing some problems outright. It would also reduce the number of available research topics, reduce the demand for funding, and threaten the careers of people whose work is built on the frameworks being removed. The bench is not cleared because clearing it has professional costs for the people who would do the clearing.

The language preserves separations that measurements have unified. Mass and inertia are one phenomenon with two names. The two names create two conceptual slots in the formalism. The two slots create the equivalence principle as an axiom. The axiom generates a research program aimed at explaining the equivalence. The research program would be unnecessary if the language used one name. The language does not change because the research program depends on the separation, and the researchers depend on the research program.

The departmental structure prevents cross-domain analysis. Particle physics, cosmology, and nuclear physics are separate departments with separate journals, separate conferences, separate funding streams, and separate career tracks. A derivation chain that starts from gauge integers in particle physics and ends at nuclear abundances in cosmology crosses three departmental boundaries. No department owns the chain. No journal is the natural home for the result. No career track rewards the work. The result falls between departments. The departments do not reorganize to accommodate it because departmental boundaries are administrative structures that serve institutional functions — hiring, funding, evaluation — independent of whether they align with the physics.

The publication system incentivizes novel frameworks over empirical cleanup. A paper proposing a new theoretical framework for quantum gravity is publishable. A paper demonstrating that an existing framework fails a specific mechanical test is less publishable. A paper clearing an unfalsified interpretation from the workbench has no publication venue because clearing the bench is not recognized as a scientific contribution. The publication incentive points toward adding to the bench, not clearing it. The bench grows.

The funding structure rewards open problems. Grant applications are organized around problems. A researcher seeking funding describes the problem, explains why it is important, proposes an approach, and requests resources. The existence of the problem is the justification for the funding. If the problem is closed, the justification disappears. The funding structure creates a financial incentive to maintain problems in an open state — to make progress toward closure without achieving closure. Complete closure terminates the funding stream. Partial progress renews it.

Each of these structural features reinforces the others. The data pipeline prevents investigation. The credential gate prevents outside perspectives. The unfalsified frameworks generate reconciliation problems. The language preserves false separations. The departments prevent cross-domain work. The publication system rewards addition over subtraction. The funding system rewards openness over closure. Remove any one feature and the others maintain the status quo. The system is stable. The problems are stable. The funding is stable. The phrase “it’s hard” is stable. Nothing changes because the architecture prevents change.

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## 13. What Closure Looks Like

Closure is not a theoretical concept. It has a working demonstration.

A reference implementation exists. It consists of a value store containing over 5,000 named, typed, versioned, machine-readable entries spanning quantum electrodynamics, electroweak physics, cosmology, Big Bang nucleosynthesis, CKM mixing, confinement, gravity, and more. Each entry carries its source — the specific paper, table, and year from which the value was obtained. Each carries its uncertainty, its significant digits, its unit, and its methodological context. No entry exists without a source. No value enters a computation without being declared.

Fifty experiments have been run against this value store. Each experiment has a machine-readable definition that declares its inputs, its derivation steps, its expected outputs, and its comparison criteria — all written before the computation runs. Each comparison is a pre-registered falsification condition: a specific prediction, a specific tolerance, and a mechanical verdict of PASS, FAIL, or INFO determined by the runner without human interpretation.

One experiment starts from two integers — 11 and 13, drawn from the gauge structure of the Standard Model — and derives primordial nuclear abundances through a chain that crosses particle physics, cosmology, and nuclear physics. The chain produces a prediction for the primordial deuterium abundance that matches spectroscopic measurements of high-redshift gas clouds to three significant digits, with a miss of 0.14% and a deviation of 0.12 standard deviations from the measurement. The primordial helium prediction falls within one standard deviation of the measurement. The baryon density hits the Planck satellite value at 727 parts per million. Seven derivations. Zero errors. Fifty-seven output values. Every one traceable.

The experiment prints its own failures. The helium digit-agreement test misses. The report says FAIL. The miss is 1.49%. The comparison was pre-registered in the experiment definition. The runner evaluated it mechanically. The failure is printed with the same precision as the successes. Nothing is hidden because nothing can be hidden — the infrastructure has no mechanism for selective reporting.

Another experiment tests a soliton-based gravity framework against every major classical test of general relativity. Mercury’s perihelion precession: predicted 42.9800 arcseconds per century, measured 42.9799, miss 2.8 parts per billion. Hulse-Taylor binary pulsar orbital decay: predicted ratio 0.999958, miss 42 parts per million. Solar surface gravitational redshift: miss 16 parts per million. GPS net daily clock correction: miss 0.35%. The speed of light derived from the Planck length divided by the Planck time: exact match, zero miss. One test — Gravity Probe A — fails at 2.47%, and the report prints FAIL. The failure is located precisely. The framework knows where it doesn't work.

Another experiment uses integer relation algorithms to scan the analytically unknown coefficients in four-loop QED calculations against a basis of 66 mathematical constants. Seventeen scans. All return null — no integer relation found. Nineteen pre-registered comparisons. Nineteen passes. The experiment expected null and got null. The search space is narrowed. These constants are not expressible as integer combinations of this particular basis. The closure is negative — the answer is "not this" — and it is a finding.

Another experiment derives cosmological matter ratios from the beta function coefficients of the Standard Model gauge groups through exact rational arithmetic. The dark matter to baryon ratio prefactor comes out as 22/13 — an exact fraction from gauge integers. Twenty-two passes, zero failures, every result an exact fraction.

This was built on a laptop. No committee. No multi-year publication cycle. No credential review. No departmental approval. No averaging pipeline. A value store, a runner, experiment definitions, and the discipline to declare inputs, pre-register comparisons, and print failures. The workbench is clean. Every item on it has a source, a test, and a verdict. The methodology is described in full in [HOWL-CULT-15-2026].

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## 14. Falsification Conditions

This paper practices what it prescribes. Each major claim is accompanied by an explicit condition under which it would be false. Each condition is observable. Each could fail. The paper commits to each and accepts falsification if it comes.

**Claim 1: The non-convergence of G is due to uninvestigated contextual dependence.** If CODATA publishes raw time-stamped measurement data from all contributing instruments with full metadata, and analysis of the raw data reveals that the non-convergence has no correlation with any contextual variable — device type, location, altitude, geology, time, solar activity, seismic state, geomagnetic conditions — then the non-convergence is not caused by uncontrolled contextual factors. The data destruction critique would retain value for transparency, but this paper's specific claim about G would have failed.

**Claim 2: Theta = 0 is a measurement, not a mystery.** If theta is measured at a nonzero value in a future experiment, the claim that the field should accept zero as the measured value is falsified. The strong CP problem would become a real problem rather than a constructed one. This paper would have been wrong about theta.

**Claim 3: The block universe is physically impossible.** If a physical mechanism for the block universe is proposed that specifies a storage substrate, an energy budget, and thermodynamic signatures, and those signatures are experimentally detected, the block universe critique fails. The concept would be promoted from unfalsifiable interpretation to testable physical theory — and it would have passed its test.

**Claim 4: The credential gate excludes scientifically valuable work.** If the credential gate is removed and a content-based evaluation system is implemented, and the system processes a sufficiently large representative sample of outsider submissions, and zero submissions survive content evaluation to the level of field engagement, then the credential system is not excluding valuable work. The gate critique would retain value for the provision of located feedback to submitters, but the claim that valuable contributions are being lost would have failed.

**Claim 5: The institutional architecture is the primary obstacle, not the difficulty of the problems.** If the structural changes described in this paper — raw data publication, content-based evaluation, workbench clearing, cross-domain infrastructure — are implemented and the open problems remain open after a sustained period with the new architecture, then the problems may genuinely be hard. The architectural critique would have failed. The problems would have survived the removal of the obstacles this paper identifies as the cause of their persistence.

**Claim 6: The universe is discrete and finite.** If a continuous mathematical formalism is shown to be the mechanism of physical reality rather than an approximation — operating at sub-Planck timescales and producing physically meaningful results below the Planck scale — then the discrete universe claim fails. The continuous mathematics would be the reality, not an effective description of a discrete substrate.

Each of these conditions is specific. Each is observable. Each could fail. The paper commits and accepts falsification if it comes.

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## 15. Closing

The universe closes every tick. Finite state. Finite update. Finite output. Move on. Each Planck time, the entire state updates. No residuals. No open items. No deferred calculations. The tick completes and the next tick begins.

The institution doesn't close anything.

The strong CP problem has been open since the 1970s. The hierarchy problem has been open since the 1970s. The cosmological constant problem has been open since the 1980s. Quantum gravity has been open since the mid-20th century. The non-convergence of G has persisted across every CODATA cycle. Dark matter has been sought for decades. None of these problems have been closed. None are on a trajectory toward closure. Each generates ongoing funding, ongoing publication, ongoing conference presentations, and ongoing assertions that the problem is hard.

The problems are not hard. The data pipeline destroys the information needed to investigate them. The credential gate excludes the perspectives needed to see them clearly. The unfalsified frameworks clutter the workbench and force false reconciliations. The language preserves separations the measurements have unified. The departments prevent the cross-domain work the problems require. The publication system rewards adding to the bench over clearing it. The funding system rewards maintaining problems over closing them.

Change the architecture and the problems become tractable. Publish the raw data and the non-convergence can be investigated. Accept measured zeros and constructed mysteries dissolve. Clear unfalsified interpretations from the workbench and the reconciliation landscape simplifies. Evaluate work on content rather than credentials and the field gains perspectives it cannot generate internally. Enable cross-domain analysis and connections between quantities in different departments become visible.

Or don't change anything. Keep the architecture. Keep the gate. Keep the cluttered bench. Keep the destroyed data. Keep the separated language. Keep the departmental boundaries. Keep the publication incentives. Keep the funding structure.

And keep saying it's hard.

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**Series cross-references (for deeper treatment of concepts introduced in this paper):**

- Statistical breadth substituting for functional depth: [HOWL-CULT-1-2026]
- The institutional gap between measurement domains: [HOWL-CULT-2-2026]
- The structural cessation of falsification practice: [HOWL-CULT-3-2026]
- What falsification structurally requires of a measurement: [HOWL-CULT-4-2026]
- Why located errors are the most valuable finding: [HOWL-CULT-5-2026]

- Publications as immutable timestamps: [HOWL-CULT-6-2026]
- Inherited enforcement of untested norms: [HOWL-CULT-7-2026]
- Working in the space between institutional departments: [HOWL-CULT-8-2026]
- Institutional structure for cross-domain work: [HOWL-CULT-9-2026]
- The structural mechanics of institutional non-commitment: [HOWL-CULT-12-2026]
- Replacing averaged snapshots with synchronized measurement: [HOWL-CULT-13-2026]
- Content-based evaluation of scientific contributions: [HOWL-CULT-14-2026]
- Structural specification for reproducible computation: [HOWL-CULT-15-2026]


[**HOWL-CULT-16-2026**] It's Hard: Why the Physics Community Will Never Close Its Open Problems. Github: https://github.com/ghowland/papers/tree/main/papers/CULT/HOWL-CULT-16-2026

[**HOWL-CULT-1-2026**] The N=1000 Fallacy. Github: https://github.com/ghowland/papers/tree/main/papers/CULT/HOV
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[**HOWL-CULT-5-2026**] Falsification as Finding. Github: https://github.com/ghowland/papers/tree/main/papers/CULT/HC
CULT-5-2026

[**HOWL-CULT-6-2026**] Immutable and Mortal. Github: https://github.com/ghowland/papers/tree/main/papers/CULT/HC
CULT-6-2026

[**HOWL-CULT-7-2026**] The Five Monkeys Problem. Github: https://github.com/ghowland/papers/tree/main/papers/CUL
CULT-7-2026

[**HOWL-CULT-8-2026**] The Omni-Domain Investigator. Github: https://github.com/ghowland/papers/tree/main/papers/C
CULT-8-2026

[**HOWL-CULT-9-2026**] The Omni-Academy. Github: https://github.com/ghowland/papers/tree/main/papers/CULT/HOV
CULT-9-2026

[**HOWL-CULT-12-2026**] Cowardice and High Status as Anti-Popper Progress Blocking.

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[**HOWL-CULT-15-2026**] Showing Your Work in 2026. Github: https://github.com/ghowland/papers/tree/main/papers/CU
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